



<http://www.pm-primer.com/wp-content/uploads/2012/04/wbs1.jpg>

WORK BREAKDOWN STRUCTURES

LECTURE 5

G.R. Labbé

WORK BREAKDOWN STRUCTURE

Definition: A Work Breakdown Structure (WBS) is a deliverable oriented grouping of project components that organizes and defines the total scope of the project.

- ❖ The WBS is the project scope broken down first into smaller manageable pieces that define major outcomes or deliverables of the project.
- ❖ Each major outcome is then further divided into smaller more manageable “chunks” or work packages.
- ❖ These work packages identify the tasks that need to be completed to produce the deliverable.
- ❖ Project Managers cannot, in fact nobody can, manage projects.
Project Managers manage pieces of projects.



WORK BREAKDOWN STRUCTURE

- ❖ The goals of a well developed WBS are:
 - ❖ Allows the project team to **logically plan** all the steps required to successfully complete the project.
 - ❖ **Collect** and **organize** all information required to complete the defined tasks.
 - ❖ Organize tasks into **smaller more manageable components** that can be assigned to the team members for completion.

A well defined WBS allows for the creation of a **WBS dictionary**. The dictionary defines the **requirements for each task** and allows for more detail to be presented.



HIERARCHICAL STRUCTURE

Note the use of a numbering sequence to identify levels.

Level 1 is defined as the Project level
(*What is the project?*)

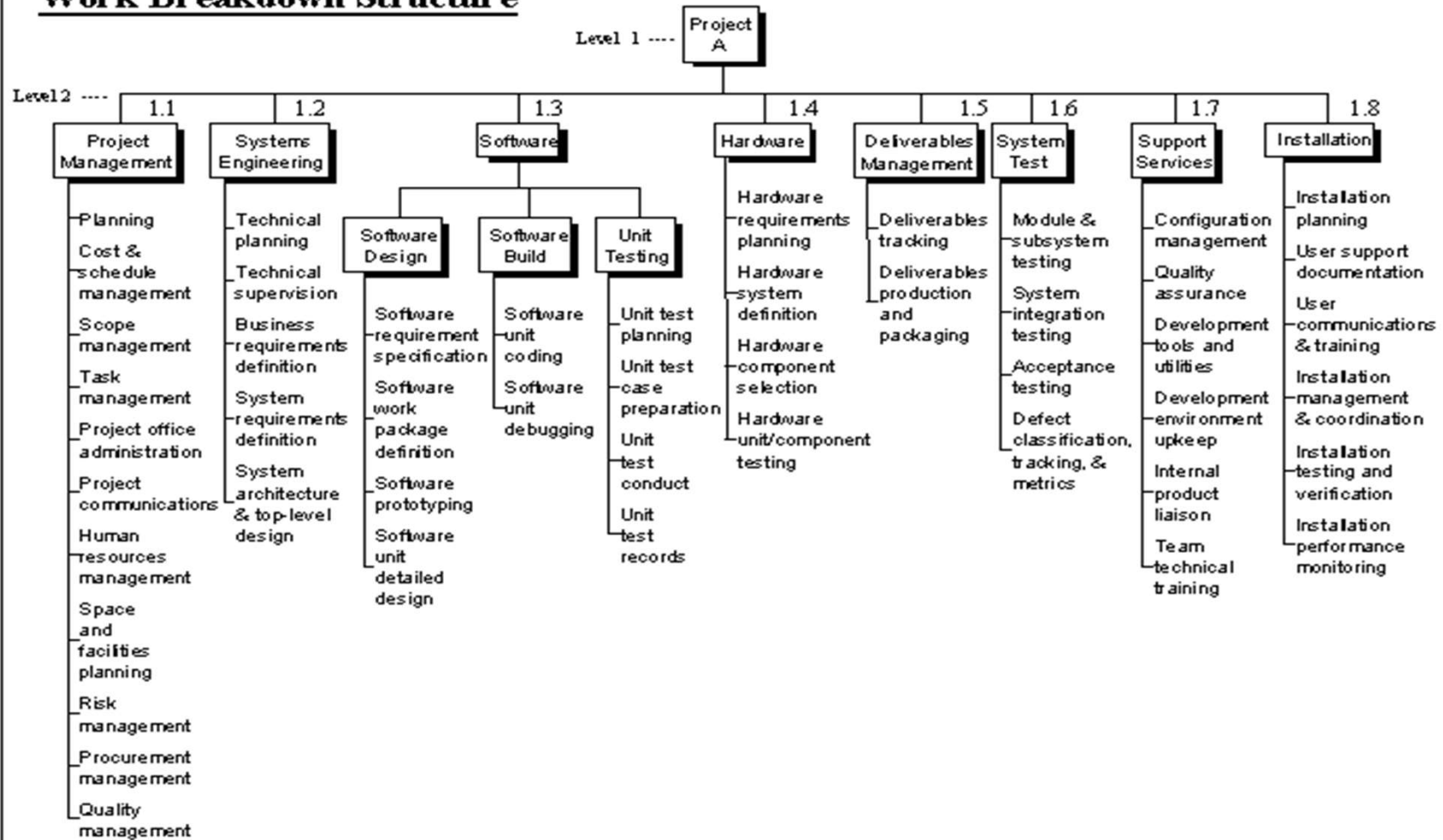
Level 2 is the Summary level. (*This is the project broken into its major components.*)

Level 3 is the start of the Work Package level. (*Depending upon the complexity of the work package there can be further levels created.*)

	Level	WBS Code	Element Name
1			
2	1	1	Widget Management System
3	2	1.1	Initiation
4	3	1.1.1	Evaluation & Recommendations
5	3	1.1.2	Develop Project Charter
6	3	1.1.3	Deliverable: Submit Project Charter
7	3	1.1.4	Project Sponsor Reviews Project Charter
8	3	1.1.5	Project Charter Signed/Approved
9	2	1.2	Planning
10	3	1.2.1	Create Preliminary Scope Statement
11	3	1.2.2	Determine Project Team
12	3	1.2.3	Project Team Kickoff Meeting
13	3	1.2.4	Develop Project Plan
14	3	1.2.5	Submit Project Plan
15	3	1.2.6	Milestone: Project Plan Approval
16	2	1.3	Execution
17	3	1.3.1	Project Kickoff Meeting
18	3	1.3.2	Verify & Validate User Requirements
19	3	1.3.3	Design System
20	3	1.3.4	Procure Hardware/Software
21	3	1.3.5	Install Development System
22	3	1.3.6	Testing Phase
23	3	1.3.7	Install Live System
24	3	1.3.8	User Training
25	3	1.3.9	Go Live
26	2	1.4	Control
27	3	1.4.1	Project Management
28	3	1.4.2	Project Status Meetings
29	3	1.4.3	Risk Management
30	3	1.4.4	Update Project Management Plan
31	2	1.5	Closeout
32	3	1.5.1	Audit Procurement
33	3	1.5.2	Document Lessons Learned
34	3	1.5.3	Update Files/Records
35	3	1.5.4	Gain Formal Acceptance
36	3	1.5.5	Archive Files/Documents

STRUCTURE VIEW

Work Breakdown Structure



http://www.hyperhot.com/pm_wbs_swhw.gif

TABULAR VIEW

Project Name	Task 1	Subtask 1.1	
			Work Package 1.1.1
			Work Package 1.1.2
		Subtask 1.2	
			Workpackage 1.2.1
	Task 2		Workpackage 1.2.2
		Subtask 2.1	
			Workpackage 2.1.1
			Workpackage 2.1.2

Each level clearly breaks the tasks down into smaller more manageable tasks.



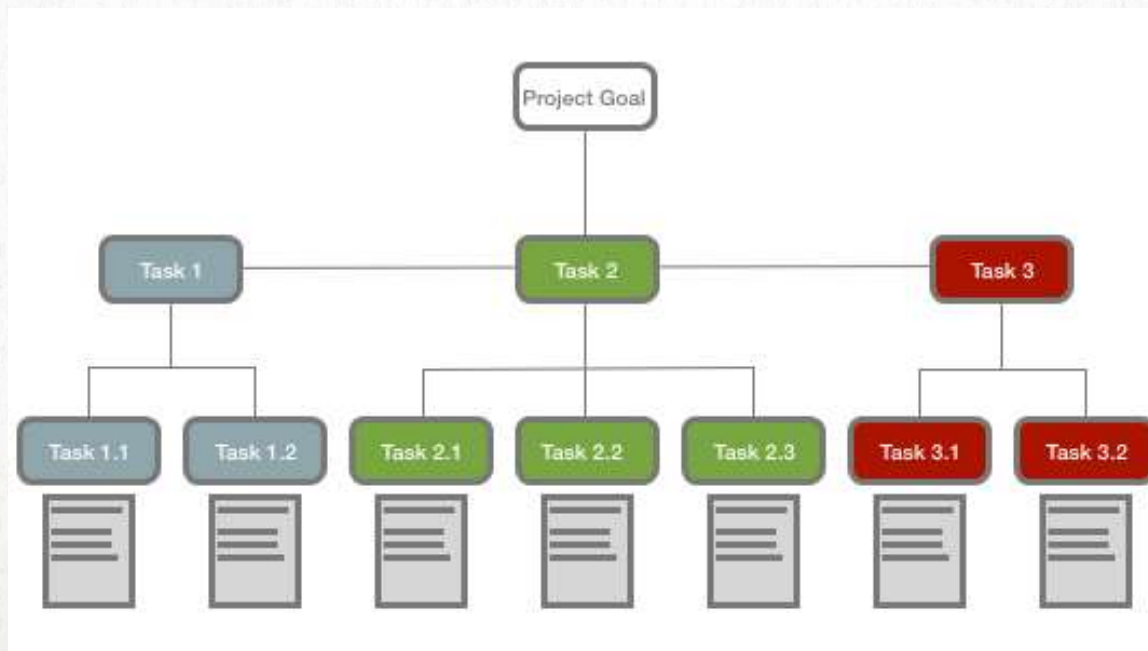
CREATING A WBS

- The WBS is organized around the primary products of the project (*or planned outcomes*) instead of the work needed to produce the products (*planned actions*).
- For example:
- ☒ Outcome-based: “Training Module Completed”
- ☐ Action-based: “Write training script,” “Record video,” “Upload to platform”
- Since the *planned outcomes* are the desired ends of the project, they form a relatively stable set of categories in which the costs of the planned actions needed to achieve them can be collected.
- A well-designed WBS makes it easy to assign each project activity to one and only one terminal element of the WBS.
- There are several **design principles** that must be followed when creating a WBS. We are going to look at these in detail in the next few slides:



DESIGN PRINCIPLES – 100% RULE

- The **100% rule** states that the WBS includes 100% of the work defined by the project scope and captures all deliverables – internal, external, interim – in terms of the work to be completed.
- The rule applies at all levels within the hierarchy: the sum of the work at the "child" level must equal 100% of the work represented by the "parent" and the WBS should not include any work that falls outside the actual scope of the project, that is, **it cannot include more than 100% of the work.**



<http://www.exceltemp.com/wp-content/uploads/2014/07/WBS.jpg>

DESIGN PRINCIPLES – MUTUALLY EXCLUSIVE

- It is important that there is **no overlap** in scope definition between different elements of a work breakdown structure. This ambiguity could result in duplicated work or miscommunications about responsibility and authority.
- Such overlap could also cause confusion regarding project cost accounting. If the WBS element names are ambiguous, a **WBS dictionary** can help clarify the distinctions between WBS elements.

WBS Dictionary			
Project Name:	WBS Name:	WBS Id:	Parent Id:
WBS Owner:		Start Date:	End Date:
WBS Detail:			
WBS Element Description:			
Acceptance Criteria:			
Assumptions:			
Resources Assigned:			
WBS dependencies:			
Cost:			
Approved by:		Date:	

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DESIGN PRINCIPLES – PLAN OUTCOMES, NOT ACTIONS

- If the work breakdown structure designer **attempts to capture any action-oriented details** in the WBS, he/she will likely include either **too many actions** or **too few actions**.
- Too many actions will exceed 100% of the parent's scope and too few will fall short of 100% of the parent's scope. The best way to adhere to the 100% rule is to define WBS elements in terms of **outcomes** or results, not actions.
- This also ensures that the WBS is not overly prescriptive of methods, allowing for greater ingenuity and creative thinking on the part of the project participants.



DESIGN PRINCIPLES – LEVEL OF DETAIL

- A **work package** at the activity level is a task that:
 - can be realistically and confidently **estimated**;
 - makes no sense practically to break down any further;
 - produces a **deliverable** which is **measurable**; and
 - forms a **unique package of work** which can be outsourced or contracted out.
- One must decide when to stop dividing work into smaller elements. This will assist in determining the duration of activities **necessary** to produce a deliverable defined by the WBS.

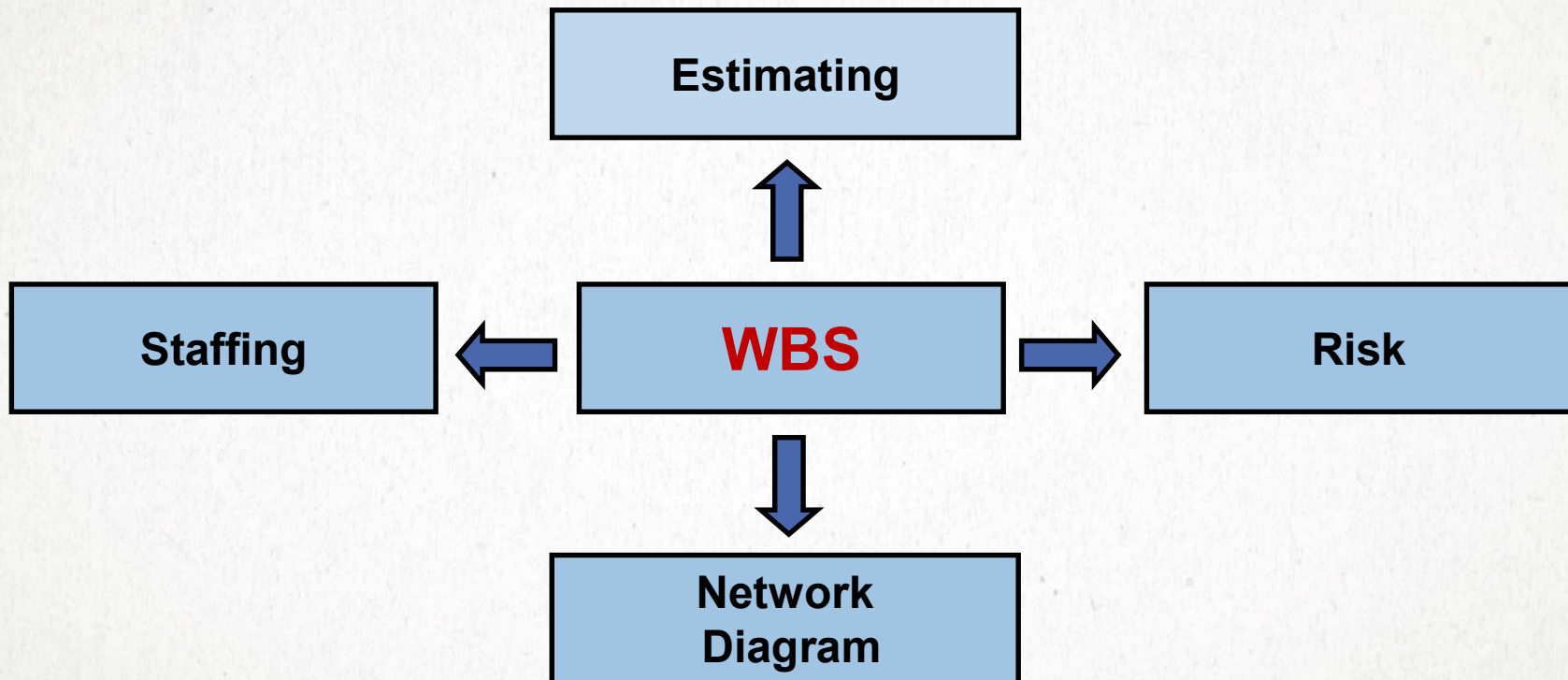


DESIGN PRINCIPLES – LEVEL OF DETAIL

- There are several **"rules of thumb"** used when determining the appropriate duration of an activity or group of activities necessary to produce a specific deliverable defined by the WBS.
 - The first is the **"80-hour rule"** which means that no single activity or group of activities at the lowest level of detail of the WBS to produce a single deliverable should be **more than 80 hours of effort**.
 - The second rule of thumb is that no activity or group of activities at the lowest level of detail of the WBS should be longer than a single reporting period. Thus, if the project team is reporting progress monthly, then no single activity or series of activities should be longer than one month long.
 - In general, no task is defined at **less than 8 hours to complete**.
 - The last one is the **"if it makes sense"** rule. Applying this rule of thumb, one can apply **"common sense"** when creating the duration of a single activity or group of activities necessary to produce a deliverable defined by the WBS.



WHY IS THE WBS SO IMPORTANT ?



WBS DICTIONARY - EXPLAINED

- Used to **control** what work is done and when;
- Used to prevent **scope creep**;
- Used to **understand** the effort for each task;
- Physically the WBS dictionary is written when the WBS work packages are posted on a wall or white board in a logical sequence relative to one another to reflect the flow of work. When they are removed in numerical order, the WBS Dictionary is generated;
- Entries in the dictionary are referred to as **task descriptions**; and,
- Frequently they are arranged on color coded work packages to clearly identify the hierarchy of the task.

WBS DICTIONARY

WBS Dictionary (<i>Task Description</i>)			
Project name: _____	Task #: _____	Date Issued: _____	Person Assigned: _____
Task Description: (<i>what work is authorized</i>)			
Quantified Objectives: (<i>See the description in the project charter</i>)			
Product Description: (<i>what is the result of the work package?</i>)			
Acceptance Criteria: (<i>how will the team member be able to check their work?</i>)			
Deliverables: (<i>See the description in the project charter</i>)			
Assumptions:			
Resources Assigned:			
Duration:			
Cost:			
Due Date:			
Interdependencies: Before this task _____ After this task _____			
Approved by: Project Manager _____ Date _____			

WBS DICTIONARY

WBS #:	1.1.1	Task:	Create Plan
Est. Level of Effort:	40 hrs	Owner:	Project Manager
Resources Needed:	Subject Matter Experts	Work Products:	MS Project Plan
Description of Task:	Development of a detailed project plan that lists all key resources, tasks, milestones, dependencies, and durations.		
Input:	• Approved Project Charter • SMEs		
Dependencies:	• Approval of Budget		
Risk:	• Changes to IT Apps plans and deliverables • IT Apps implementation releases, which conflict with implementation		
WBS #:	1.1.2	Work Item:	Make Budget
Est. Level of Effort:	16 hrs	Owner:	Project Manager
Resources Needed:	CFO, CIO, Executive Sponsor	Work Products:	ITPR
Description of Task:	Development and documentation of the project budget based on plan and resources.		
Input:	• Approved Project Charter • SMEs		
Dependencies:	• Approval of Project Charter		
Risk:	• Changes to IT Apps plans and deliverables • IT Apps implementation releases which conflict with implementation		

WBS Dictionary - Table Format Example

doit.maryland.gov/sdlc/formservertemplates/wbs.doc

WBS DICTIONARY EXERCISE

- In your group, take one task from your WBS that you created earlier and create a WBS dictionary for it.

WORK AUTHORIZATION

Once the scope has been decomposed into useful work packages, the next natural step is to get **formal agreement between stakeholders** regarding project content known as the **work authorization**. **If the client is external, this likely includes the contract Sign-Off.**

Project management contracts to be valid require and consist of at least:

1. Possessing the legal right to contract;
2. Being of legal age to contract (*18yo in Ontario and Québec and 19 or 21yo in other jurisdictions in Canada*);
3. Being of sound mind at the time of signing;
4. Valid and Legal Consideration for both parties;
5. A legal scope of work; and,
6. Offer and acceptance.



WORK AUTHORIZATION

Project management **contracts** also likely contain many of the following *(but not limited to)*:

- ❖ Excusable delays;
- ❖ Allowable costs;
- ❖ Statements of liquidated damages *(non-performance clauses)*;
- ❖ Criteria for Inspection;
- ❖ Responsibility for correction of defects; and,
- ❖ Dispute resolution mechanisms.